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DILONG OBC DC/DC INTEGRATED - COMMUNICATION PROTOCOL V1.4

BMS/CHARGER NODE

CAN Bus Communication Protocol Specification

Application range: Newly ordered OBC DC/DC integrated charger after 2019.12.01

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Version	Content changed	Principle	Finished Date
V1.0	Drafting	Wang Bing	2019．01．19
V1.1	Charger is added heating mode, DC/DC is added parallel function	Wang Bing	2019．01．21
V1.2	Adding the baud rate explaination and DC part data format description	Li Qiang	2019．03．07
V1.3	Adding explanation of Baud Rate of 125KBPS	Li Qiang	2019.04.10
V1.4	Modify the send ID. Adding charger temperature information	Li Qiang	2019.11.25

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1. Network structure and hardware specification

1.1. CAN network adopts bus topology, the nodes on the network include: Battery Management System, Charger Control System, DCDC and VCU four nodes, each node is connected to the bus through the shielded twisted pair line.

1.2. CAN information frame format standard: 2.0B 29bit Identifier extended frame ID，physical layer ISO11898，SAE J1939，120Ω Matching resistance, Current symmetric drive.

1.3. *By default, this protocol defaults without termination resistors.*

2. Communication conventions of CAN network

CAN network communication rate is 250Kbps.

3. Definition of communication mode and identifier

3.1. Communication mode: communication between nodes of the network runs according to the following ways: after each node controller initialization runs, it send data to the bus in a broadcast manner according to a certain period. The diagnostic message sends data to the bus by event triggering. All data are in the following format: motorola format.

3.2. Identifier definition (in the format of SAE J1939): the following is the allocation table for the 29 identifier (ID) in CAN:

IDENTIFIER 11BITS											R R	D E	IDENTIFIER EXTENSION 18BITS																	
PRIORITY				P	PDU FORMAT(PF)						R R	D E	PF		PDU SPECIFIC(PS)								SOURCE ADDRESS(SA)							
8	7	6	5	4	3	2	1	0	9	8			7	6	5	4	3	2	1	0										

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Among them, the priority is 3 bits, which can have 8 priorities, from the highest 0 to the lowest 7, and the charging response information, charging status information, charging stage alarm information priority is 5, and the default priority of other information is 6.

R is generally determined to be 0.

DP was tentatively determined to be 0.

The 8 - bit PF is the code of the message;

The 8 bit PS is the target address or group extension;

The 8 bit SA is the source address for sending the message;

a. Each node in the access network has the name and address, and the name is used to identify the function of the node and to arbitrate the address, and the address is used for the data communication of the node.

b. Each node has at least one function, and there may be multiple nodes that have the same function, and one node may have multiple functions.

4. COMMUNICATION NETWORK MESSAGE PROTOCOL

Protocol description:

4.1. The order of Byte and Bit is as follows:

CAN information frame sequence								
High←-------Bit------→Low								
Bit No.	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0

Low Byte High	Byte 0	07	06	05	04	03	02	01	00
	Byte 1	17	16	15	14	13	12	11	10
	Byte 2	27	26	25	24	23	22	21	20

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	Byte 3	37	36	35	34	33	32	31	30
	Byte 4	47	46	45	44	43	42	41	40
	Byte 5	57	56	55	54	53	52	51	50
	Byte 6	67	66	65	64	63	62	61	60
	Byte 7	77	76	75	74	73	72	71	70

4.2. The data content of more than 1 Byte / 8 bit words is agreed to be sent and received in Motorola format. The data format is as follows: if the data span 2 Byte, the data will be placed in the Byte position, and the data will be placed at the larger position of Byte(Send in sequence: High bytes at the front, low bytes at the back).

4.3. According to the SAE J1939 protocol, the unused data byte value is 0xFF, and the undefined bits are sent by 1. (no compulsory regulation can be 1 or 0)

5. NODE ADDRESS ALLOCATION TABLE

Node name	Address	Note
Battery Management System (BMS)	0x180 6E5F4	Send and accept
Charger Control System (CCS1)	0x18FF50E5	Send and accept
Charger Control System (CCS2)	0x18000001	Send and accept
Charger Control System (CCS3)	0x18000002	Send and accept
DCDC Control System (DCDC)	0x18nmE5F5	Send and accept
Vehicle Control Unit (VCU)	0x18mnF5E5	Send and accept

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6. MESSAGE INFORMATION

Message code	Message description	Sending node	Receiving node	Priority	Data bytes	Refresh rate(ms)	29 bit identifier
BGS1	Battery state	BMS	Charger	6	8	1000	0x1806E5F4
CCS1	Charger state	Charger	Broadcast	6	8	1000	0x18FF50E5
CCS2	Internal flow sharing information of the charger 1	Charger	Charger	7	8	200	0x18000001
CCS3	Internal flow sharing information of the charger 2	Charger	Charger	7	8	200	0x18000002
DCDC	DCDC state information	DCDC	VCU	6	8	200	0x18nmE5F5
VCU	Charger state	VCU	DCDC	6	8	200	0x18mnF5E5

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Charger part:

Battery management system sending message Message code: BGS1

Battery management system sending—Charger control system receiving

Message ID	18 06 E5 F4		Message length		8		Period	1000ms
P6	R0		DP 0		PF6			
Message data	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	
Byte0	Maximum allowable charge end voltage high bytes (0.1V/bit offset: 0 Cases: the corresponding voltage of Vset=3201 is 320.1V)							
Byte1	Maximum allowable charge end voltage and low bytes (0.1V/bit offset: 0 Cases: the corresponding voltage of Vset=3201 is 320.1V)							
Byte2	Maximum allowable charge current high bytes (0.1A/bit offset: 0 Cases: the corresponding current of Iset=582 is 58.2A)							
Byte3	Maximum allowable charge current low bytes (0.1A/bit offset: 0 Cases: the corresponding current of Iset=582 is 58.2A)							
Byte4	0: The charger starts to recharge 1: Battery protection, charger shut out (If with CCCP function, send this command to keep S2 closed) 2: Charging is complete (if with CCCP function, sending this command will disconnect S2)							
Byte5	0: Battery mode 1: Heating mode							
Byte6	Retain							
Byte7	Retain							

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Charger control system sending message Message code: CCS1

Message ID	18 FF 50 E5		Message length		8		Period		1000ms
P6	R0		DP0		PF FF				
Message data	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	
Byte0	Output voltage high byte (0.1V/bit offset: 0 Cases: the corresponding voltage of Vset=3201 is 320.1V)								
Byte1	Output voltage low byte (0.1V/bit offset: 0 Cases: the corresponding voltage of Vset=3201 is 320.1V)								
Byte2	Output current high byte (0.1A/bit offset: 0 Cases: the corresponding current of Iset=582 is 58.2A The highest Bit mark symbol, 0 for charging, 1 for discharge)								
Byte3	Output current low byte (0.1A/bit offset: 0 Cases: the corresponding current of Iset=582 is 58.2A The highest Bit mark symbol, 0 for charging, 1 for discharge)								
Byte4	State sign: STATUS (see Schedule 1)								
Byte5	Charging pilot alarm (see Schedule 2). **If the charger does not have CCCP function, ignore this byte signal.**								
Byte6	Charger internal temperature (accuracy: 1℃/bit, offset: -40℃)								
Byte7	Retain								

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Schedule 1 Status flag STATUS Description

STATUS	Identification	Describe
Bit0	Hardware failure	0: normal 1: Hardware failure
Bit1	Charger temperature	0: normal 1: Charger temperature is too high protection
Bit2	Input voltage	0: Input voltage is normal 1: Input voltage error, the charger stops working
Bit3	Start state	0: Charger detects the battery voltage into the start-up state 1: Off (to prevent battery reverse connection)
Bit4	Communication status	0: Communication is normal 1: Communication reception timeout
Bit5		
Bit6		
Bit7		

Schedule 2 Charge Boot Status Alarm Description

STATUS	Identification	Describe
Bit0	S2 switch status	0: Disconnect 1: close
Bit1	Electronic lock status	0: Unlock 1: Lock 2: Fault 3: No electronic lock
Bit2		
Bit3	Charging socket over temperature alarm	0: Normal 1: Alarm
Bit4	CC status alarm	0: Normal 1: Alarm
Bit5	CP status alarm	0: Normal 1: Alarm
Bit6		
Bit7		

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Way of working:

1. BMS fixed interval time 1S send control information (BGS1) to the charger, the charger receives the information according to the message data voltage and current settings to work. If 5 seconds can not receive the message, then enter the communication error state, turn off the output.

2. The charger sends a broadcast message every 1S (CCS1), indicating that the meter can display the charger status based on the message.

Note: This protocol default with no terminating resistor. The CCS2 and CCS3 messages are used for internal current sharing within the charger, If in use, CCS2 and CCS3 messages and vehicle ID allocation conflict or have special requirements, please contact our technical staff.

If the version is 500Kbps, the product model number is added "C"; If the version is 125Kbps, the product model number is added "B" ("DA5KM37-362C-14C" stands for 250Kbps; "DA5KM37-362CA-14CA" stands for Baud rate 500Kbps;"DA5KM37-362CB-14CB" stands for Baud rate 125Kbps)

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DCDC part:

1. Overview

This protocol uses CAN2.0B extended frame communication, communication cycle 200ms, communication speed of 250kbps, no termination matching resistance, the default 29-bit ID, as shown in the following table:

NO.	ID	Type	Period(ms)	Size(byte)	VMS	DCDC
1	0X18mnF5E5	P	200	8	T	R
2	0X18nmE5F5	P	200	8	R	T

Note:

1. P means the cycle to send
2. T means sending node
3. R means receiving node
4. VCU means the node that communicates directly with the DC-DC converter
5. m in ID "0X18mnF5E5" and "0X18nmE5F5" means the power supply type

Output voltage range	M value
0V≦V＜16V	0
16V≦V＜32V	1
32V≦V＜52V	2
52V≦V＜62V	3
62V≦V＜78V	4
78V≦V＜100V	5

6. n in ID "0X182nF5E5" and "0X18n6E5F5" means stands for the meaning of the nth machine, starting from the 0th machine by default. In the same group, the n of "0X182nF5E5" and "0X18n6E5F5" must be equal, and the range of n is 0≦n≦F.

Note: For example, the ID of the VCU of the 0th machine is “0X18m0F5E5”, and the ID of the DC/DC is “0X180mE5F5”
The ID of the No. 1 VCU is “0X18m1F5E5”, and the ID of DC/DC is “0X181mE5F5”

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The ID of the No. 2 VCU is "0X18m2F5E5", and the ID of DC/DC is "0X182mE5F5"
And so on...
The ID of the No. n VCU is "0X18mnF5E5", and the ID of the DC/DC is "0X18nmE5F5"
Up to 16 power supplies can be connected in parallel under the same type of power supplies.

2. FORMAT DESCRIPTION

CAN specific information bit definitions as shown in the following table:

	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
Byte0	7	6	5	4	3	2	1	0
Byte1	15	14	13	12	11	10	9	8
Byte2	23	22	21	20	19	18	17	16
Byte3	31	30	29	28	27	26	25	24
Byte4	39	38	37	36	35	34	33	32
Byte5	47	46	45	44	43	42	41	40
Byte6	55	54	53	52	51	50	49	48
Byte7	63	62	61	60	59	58	57	56

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3. VCU MESSAGE PROTOCOL

OUT	IN	CAN ID	Cycle (ms)
VCU	DC/DC	0X18mnF5E5	200

Data		
Location	Data name	
BYTE0	Reserve	
BYTE1	Switch on/off control	Bit0: 1: Power on command 0: Shutdown command
BYTE2	Set voltage low Byte (low 8 bit)	0.1V/bit Offset: 0 e.g Vset = 140(BYTE2+ BYTE3*256), its corresponding 14.0V
BYTE3	Set voltage high Byte (high 8 bit)	
BYTE4	Set current low Byte (low 8 bit)	0.1A/bit Offset: 0 e.g Iset = 1000(BYTE4+ BYTE5*256), its corresponding 100A
BYTE5	Set current high Byte (high 8 bit)	
BYTE6	Reserve	
BYTE7	Reserve	

Note:

1. Bit0 of Byte1 is 1 means on, 0 means off.

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4. DCDC MESSAGE PROTOCOL

OUT	IN	CAN ID	Cycle（ms）
DC/DC	VCU	0X18nmE5F5	200
Data			
Location	Data name		
BYTE0	Reserve		
BYTE1	Fault alarm low Byte		Bit0：Over temperature shutdown 0: normal, 1: fault Bit1：Output over-current alarm 0: normal, 1: fault Bit2：Output over-voltage alarm 0: normal, 1: fault Bit3：Output under voltage alarm 0: normal, 1: fault Bit4：Input over-voltage alarm 0: normal, 1: fault Bit5：Input under voltage alarm 0: normal, 1: fault Bit6：Output short circuit protection 0: normal, 1: fault Bit7：Internal fault alarm 0: normal, 1: fault
BYTE2	Fault alarm high Byte		Bit0：Communication fault alarm 0: normal, 1: fault Bit1：Reserve Fault location 0: normal, 1: fault Bit2：Reserve Fault location 0: normal, 1: fault Bit3：Reserve Fault location 0: normal, 1: fault Bit4：Reserve Fault location 0: normal, 1: fault Bit5：Reserve Fault location 0: normal, 1: fault Bit6：Reserve Fault location 0: normal, 1: fault Bit7：Reserve Fault location 0: normal, 1: fault
BYTE3	Output current low Byte (low 8 bit)		0.1A/bit Offset：0 e.g Iset = 1000，its corresponding 100A
BYTE4	Output current high Byte (high 8 bit)		
BYTE5	Output end voltage		m<2 0.2V/bit Offset：0 m≧2 1V/bit Offset：0
BYTE6	Working status		0: work 1: stop
BYTE7	Shell temperature		1V/bit Offset：-40 e.g Temp = 40，its corresponding 0℃

Note:

1. Byte1 and Byte2 contains all the items: 0 means normal, 1 means failure.
2. The input over-voltage and under-voltage fault DC-DC does not need to be restarted. When the input voltage returns to the normal range (refer to electrical specifications) DC-DC automatically resumes work.
3. Temperature failure DC-DC does not need to be restarted, when the temperature returned to normal range (reference electrical indicators) DC-DC automatically resume work.
4. The output short circuit, output over-voltage and under-voltage faults do not need to be restarted, and DC-DC automatically resumes operation after troubleshooting.
5. There is a 40 offset in the temperature calculation, uploading 0° C at -40° C and uploading 90° C at 50° C; that is, the uploaded value minus 40 is the normal temperature value.
6. Output voltage when m<2, accuracy is 0.2V/bit; when m≧2, accuracy is 1V/bit.

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